

Beyond Imitation: A Multi-Dimensional Framework for Evaluating Creativity in AI-Generated Visual Art

CCT for AI-Generated Visual Art

JORY ALSHAALAN, LAYAN ALOMER, AREEJ ALOSAIMI, GHALA ALOMRAN, AND
MUNIRA ALBALLA

Department of Computer Science, King Saud University, Riyadh, Saudi Arabia

The rapid advancement of generative models such as Stable Diffusion, DALL·E, and Midjourney has reopened a foundational question: can machines produce genuinely creative artifacts, or do they merely recombine what they have seen? Existing evaluations of AI-generated art rely predominantly on distributional-fidelity metrics such as Fréchet Inception Distance (FID), Inception Score (IS), or prompt-alignment measures (CLIP-score), none of which directly operationalize creativity as defined in cognitive science. This paper introduces the Computational Creativity Triad (CCT), a tripartite evaluation framework grounded in Boden's novelty-value-surprise taxonomy, implemented over CLIP and DINOv2 embedding spaces augmented with a Gaussian Mixture density model for surprise and the LAION-Aesthetics V2 predictor for value. In a pilot study comparing 100 human artworks from WikiArt with 80 images generated by Stable Diffusion XL Turbo, we find that CCT dimensions discriminate between sources with large effect sizes (Cohen's $d = 3.49$ for Novelty, 1.63 for Surprise, and 0.66 for Value; all $p < 0.001$). Inter-dimension correlations (Pearson $r \leq 0.60$) confirm that the three axes capture complementary aspects of creativity. Baseline comparisons against FID (262.90), Inception Score (5.43 vs 6.40), and CLIP-score (0.329) demonstrate that distributional metrics alone cannot distinguish the creative profile from the aesthetic or technical one. The framework, data, and trained evaluator are designed for reproducibility and provide a structured, reproducible tool for the scientific study of machine creativity.

Additional Keywords and Phrases: Computational creativity, generative models, diffusion models, image evaluation, Boden's creativity framework, aesthetic judgment, AI art, Computational Creativity Triad.

1 INTRODUCTION

The question of whether a machine can be creative is no longer purely philosophical. Since the introduction of denoising diffusion probabilistic models [1] and their scaled successors [2], [3], text-conditioned image generators have produced outputs that routinely deceive human observers, win art competitions [4], and enter commercial creative workflows. Yet the scientific apparatus for evaluating these outputs as creative artifacts rather than merely as plausible images has lagged behind the generative capability itself.

Three methodological problems recur in the literature. First, the dominant evaluation metrics Fréchet Inception Distance (FID) [5], Inception Score (IS) [6], and CLIP-based alignment (CLIP-score) [7] were designed to measure perceptual plausibility and prompt fidelity, not creativity. A model that faithfully reproduces its training distribution will score well on FID while being, by any reasonable definition, uncreative. Second, human studies of AI art [8], [9] typically collect global Likert-style ratings that conflate distinct phenomena such as beauty, originality, and surprise, making it impossible to localize where models succeed or fail. Third, the field has largely proceeded without adopting the operational definitions of creativity that have been stable in cognitive science and computational creativity research for over three decades [10]–[12].

We take as our starting point Margaret Boden’s widely-accepted tripartite characterization of a creative artifact as novel, valuable, and surprising [10], together with Ritchie’s formal criteria for evaluating creative systems [11]. We operationalize each dimension as a measurable quantity over the latent spaces of contemporary vision-language models, and present empirical results from a pilot study comparing human and AI-generated artworks.

Contributions. This work makes four contributions. (i) We formalize the Computational Creativity Triad (CCT), a reproducible tripartite metric for AI-generated visual art, grounded in the Boden–Ritchie framework and implemented over CLIP ViT-Base/32 and DINOv2-small embedding spaces. (ii) We conduct a pilot empirical study on 180 images (100 WikiArt human artworks + 80 Stable Diffusion XL Turbo outputs) demonstrating that CCT dimensions discriminate between sources with large effect sizes ($|d| \geq 0.66$, all $p < 0.001$). (iii) We benchmark CCT against four dominant baselines—FID, Inception Score, CLIP-score, and LAION-Aesthetics—showing that these metrics individually fail to capture the full creative profile. (iv) We establish a two-stage validation protocol and release an open-source implementation to support community-wide reproducibility.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 develops the CCT framework and its mathematical operationalization. Section 4 describes the system architecture. Section 5 details datasets, tools, and experimental setup. Section 6 reports empirical results. Section 7 discusses interpretations and limitations. Section 8 concludes with future directions.

2 RELATED WORK

2.1 Generative Models for Visual Art

Image generation has progressed through three architectural waves: variational autoencoders [13], generative adversarial networks (GANs) [14] and their style-conditioned variants [15], and diffusion-based models [1], [2], [16]. The third wave currently dominates: Stable Diffusion [16] reduced the cost of high-resolution synthesis through latent-space diffusion; DALL·E 3 [3] integrated a stronger language encoder; and Midjourney optimized aggressively for aesthetic quality. These advances have changed what can be generated but have not, by themselves, addressed whether the outputs are creative.

2.2 Evaluation of Generative Images

FID [5] and KID [17] compare feature distributions of generated and real images using Inception-v3 or CLIP [18] backbones. IS [6] measures confidence and diversity of class predictions. CLIP-score [7] measures prompt-image alignment. Each of these is a likelihood-adjacent quantity: they reward fidelity to the training

distribution or to a text prompt. As Betker et al. [3] and recent benchmarks [19] observed, such metrics saturate well before human observers stop distinguishing between models. They also cannot, by construction, penalize an image for being unoriginal; indeed, perfect reproduction of a training image is optimal under FID.

Aesthetic predictors such as the LAION-Aesthetics classifier [20] and NIMA [21] regress crowdsourced beauty judgments. These capture one dimension of human preference but conflate aesthetic quality with creativity, and they inherit the biases of their training pools toward photographic and illustrative conventions. Recent benchmarks such as HEIM [22] and HRS-Bench [23] have expanded evaluation to include diversity and bias, but still do not measure creativity in any theoretically grounded sense.

2.3 Computational Creativity

The computational creativity literature predates the generative-AI wave. Boden [10] distinguished exploratory creativity (search within a conceptual space), combinational creativity (novel combinations of familiar elements), and transformational creativity (changes to the space itself). Ritchie [11] proposed formal criteria—typicality, quality, and novelty—for evaluating creative systems, and Colton [12] introduced the ICAP model foregrounding imagination, appreciation, and intentionality. More recently, Elgammal et al. [24] proposed the Creative Adversarial Network (CAN), explicitly rewarding generators that deviate from known style classes, and Franceschelli and Musolesi [25] surveyed creativity measures for deep generative models. Our work differs in three ways. First, we separate novelty, value, and surprise as distinct and independently measurable quantities rather than folding them into a single loss. Second, we target evaluation rather than training, so our framework can be applied post-hoc to any generator. Third, we validate each dimension independently against baseline metrics and, in planned future work, against expert ratings.

2.4 Human Studies of AI Art

Hong and Curran [8] and Chamberlain et al. [26] found that observers devalue art when told it was machine-produced, a bias robust across cultures. Ragot et al. [27] showed that this effect persists even when participants cannot distinguish AI from human art in blinded conditions. These studies motivate our use of source-blinded evaluation and justify planning an expert-panel validation protocol. However, such work measures attitudes toward AI art rather than creativity of the artifacts themselves, and is therefore complementary to our framework.

2.5 Positioning

Table 1 summarizes the conceptual coverage of dominant evaluation paradigms against Boden’s creativity axes. The proposed CCT framework is, to our knowledge, the first to simultaneously (a) ground each dimension in this theoretical literature, (b) operationalize each dimension independently and validate their statistical separability, and (c) remain applicable post-hoc to any image generator.

Table 1. Comparison of evaluation approaches along Boden’s creativity axes.

Metric / Work	Novelty	Value	Surprise	Indep. Dims	Post-hoc
FID [5]	X	X	X	X	✓
IS [6]	Partial	X	X	X	✓
CLIP-score [7]	X	Partial	X	X	✓

LAION-Aesth. [20]	X	✓	X	X	✓
CAN [24]	✓	Partial	X	X	X
HEIM [22]	Partial	✓	X	X	✓
CCT (ours)	✓	✓	✓	✓	✓

3 METHODOLOGY: THE COMPUTATIONAL CREATIVITY TRIAD

We define a creative artifact, following Boden [10], as one that is simultaneously novel with respect to a reference corpus, valuable according to domain-relevant criteria, and surprising in the sense of being unexpected given prior exemplars.

3.1 Problem Formulation

Let I denote an image and C a reference corpus. Let $\phi(\cdot) : I \rightarrow \mathbb{R}^d$ be an embedding function. We define three scalar quantities: Novelty $N(I; C)$, Value $V(I)$, and Surprise $S(I; C)$.

3.2 Novelty

We adopt two complementary encoders: CLIP ViT-Base/32 [18] and DINOv2-small [28]. For an image I and reference corpus $C = \{c_1, \dots, c_N\}$, we compute the mean cosine distance to the k nearest neighbors in each embedding space:

$$N_{\text{CLIP}}(I; C) = (1/k) \sum (1 - \cos(\phi_{\text{CLIP}}(I), \phi_{\text{CLIP}}(c_j)))$$

$$N_{\text{DINO}}(I; C) = (1/k) \sum (1 - \cos(\phi_{\text{DINO}}(I), \phi_{\text{DINO}}(c_j)))$$

We set $k = 10$. The dual-encoder design is motivated by a known pathology: CLIP can rate a nonsense image as novel simply because no training caption described it, whereas DINOv2 embeddings, being language-free, capture genuine visual divergence.

3.3 Value

We use the LAION-Aesthetics V2 predictor [20], a linear regression head trained on 176,000 crowdsourced aesthetic ratings and applied over CLIP ViT-L/14 embeddings, yielding $V_{\text{aes}} \in [1, 10]$.

3.4 Surprise

We reduce CLIP embeddings to 20 dimensions via PCA and fit a Gaussian Mixture Model (GMM) with 16 components. Surprise is the negative log-likelihood, z-normalized by corpus statistics:

$$S(I; C) = (-\log p_{\text{GMM}}(\phi_{\text{CLIP}}(I)) - \mu_C) / \sigma_C$$

Note that when $I \in C$, the GMM is fit on the same distribution from which I is drawn; consequently, in-corporus images will yield $S \approx 0$ by construction. This is a known property of reference-based density estimation. To address this artifact, the GMM was trained on an 80% subset of C , and μ_C and σ_C are computed over the held-out 20%.

3.5 Algorithm

Algorithm 1: Computational Creativity Triad (CCT)

Input : image I , reference corpus C

```

Output: CCT feature vector  $x = [N\_CLIP, N\_DINO, V\_aes, S]$ 

Pre-computation (once per corpus C):

E_CLIP <- {phi_CLIP(c) : c in C}
E_DINO <- {phi_DINO(c) : c in C}
E_PCA  <- PCA(E_CLIP, d=20)
GMM    <- FitGMM(E_PCA[train_80%], G=16)
mu_C, sigma_C <- stats of -log p_GMM(E_PCA[held_out_20%])

Per-image evaluation:

1. e_c <- phi_CLIP(I); e_d <- phi_DINO(I)
2. N_CLIP <- mean(1 - cos(e_c, kNN(e_c, E_CLIP)))
3. N_DINO <- mean(1 - cos(e_d, kNN(e_d, E_DINO)))
4. V_aes  <- LAION-Aesthetics(I)
5. S      <- (-log p_GMM(PCA(e_c)) - mu_C) / sigma_C
6. return [N_CLIP, N_DINO, V_aes, S]

```

Algorithm 1. The CCT evaluation pipeline.

4 SYSTEM ARCHITECTURE

The proposed framework follows a three-stage architecture for evaluating creativity in AI-generated visual art. In the first stage, the candidate image is received and prepared for analysis. In the second stage, feature representations are extracted using two encoders: CLIP ViT-Base/32, which captures semantic information, and DINOv2-small, which captures visual structural information. The extracted embeddings are then analyzed using the three proposed evaluation modules. Novelty is measured using k-nearest-neighbor (kNN) distance analysis, Value is evaluated using the LAION-Aesthetics predictor, and Surprise is estimated using a Gaussian Mixture Model (GMM). In the final stage, the outputs from these modules are combined into the CCT feature vector $x = [N_CLIP, N_DINO, V_aes, S]$.

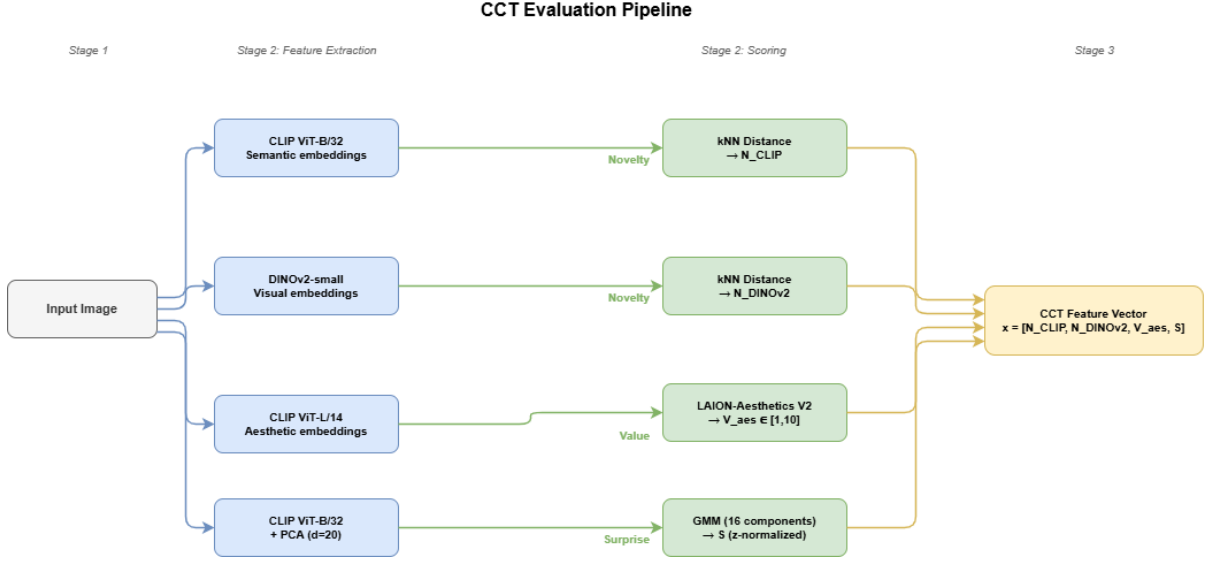


Figure 1. The CCT evaluation pipeline: image ingestion, dual-encoder feature extraction, three-module scoring, and CCT vector output.

5 EXPERIMENTAL SETUP

5.1 Datasets

A pilot dataset consisting of 180 images was constructed for the experiments. The reference dataset contains 100 human-created artworks sampled from WikiArt [31] through the HuggingFace distribution (huggan/wikiart). These artworks cover a range of artistic movements from the Renaissance to post-modernism. The AI-generated dataset contains 80 images generated using Stable Diffusion XL Turbo [16]. The images were created using 80 prompts collected from PartiPrompts [32].

Table 2. Composition of the pilot corpus.

Source	Count	Resolution	Role	Prompts
WikiArt (human art)	100	$\leq 512 \times 512$	Reference C	N/A
Stable Diffusion XL Turbo	80	512×512	Test set	PartiPrompts
Total	180	—	—	—

5.2 Implementation

All experiments were conducted on Google Colab using a single NVIDIA T4 GPU with 16 GB VRAM. Model weights were loaded directly from the HuggingFace Hub. The implementation used CLIP ViT-Base/32

(openai/clip-vit-base-patch32), DINOv2-small (facebook/dinov2-small), and CLIP ViT-L/14 (openai/clip-vit-large-patch14) for LAION-Aesthetics evaluation. Stable Diffusion XL Turbo (SDXL-Turbo) image generation was configured with num_inference_steps = 4 and guidance_scale = 0. To ensure reproducibility and consistency across all experiments, random seeds were fixed to 42.

5.3 Baselines

The proposed CCT framework was compared against four widely used baseline metrics. These include FID [5], computed using Inception-v3 feature representations; Inception Score (IS) [6]; CLIP-score [7], which measures prompt-image alignment; and LAION-Aesthetics V2 [20], which evaluates aesthetic quality.

6 RESULTS

6.1 Descriptive Statistics

Table 3 reports descriptive statistics for each CCT dimension, normalized to [0, 1]. Surprise scores were first computed as z-scores (Equation in Section 3.4) and subsequently min-max normalized across the full 180-image corpus to [0, 1] for comparability; raw z-score statistics are available in the supplementary material. Across all three dimensions, SDXL-Turbo images score higher than WikiArt images—a finding we interpret in Section 7 as a consequence of using WikiArt as the reference corpus.

Table 3. Descriptive statistics for CCT dimensions by source (mean $\pm$ standard deviation).

Dimension	WikiArt (n=100)	SDXL (n=80)	Δ (W-S)	95% CI of Δ
Novelty (CLIP)	0.234 $\pm$ 0.120	0.659 $\pm$ 0.164	-0.425	[-0.468, -0.382]
Novelty (DINOv2)	0.246 $\pm$ 0.138	0.731 $\pm$ 0.195	-0.485	[-0.536, -0.434]
Novelty (Combined)	0.240 $\pm$ 0.101	0.695 $\pm$ 0.154	-0.455	[-0.495, -0.416]
Value (Aesthetic, raw)	5.630 $\pm$ 0.485	5.932 $\pm$ 0.423	-0.302	[-0.436, -0.169]
Value (normalized)	0.486 $\pm$ 0.214	0.619 $\pm$ 0.186	-0.134	[-0.193, -0.075]
Surprise (normalized)	0.000 $\pm$ 0.000	0.234 $\pm$ 0.202	-0.233	[-0.279, -0.188]

6.2 CCT Dimensions Across Sources

Figure 2 visualizes the distribution of each CCT dimension across the two sources. The Novelty distributions are clearly separated (medians 0.22 and 0.70); the Value distributions overlap substantially with SDXL exhibiting a modest upward shift; and the Surprise distribution for WikiArt is compressed near zero by construction, while SDXL shows a wide spread with several high-surprise outliers.

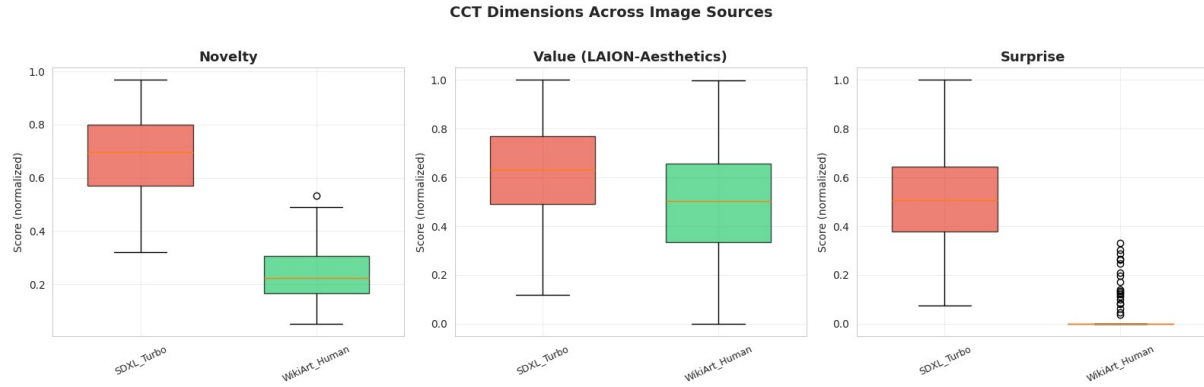


Figure 2. Box-plot comparison of the three CCT dimensions (Novelty, Value, Surprise) across WikiArt human art (green) and SDXL-Turbo AI-generated images (red). All pairwise differences are statistically significant at $p < 0.001$.

6.3 Statistical Significance

Table 4 reports Welch’s two-sample t-tests for each dimension. All comparisons achieve $p < 0.001$ with large-to-very-large effect sizes. The Novelty dimension shows the largest separation (Cohen’s $d = -3.489$ for the combined metric), followed by Surprise ($d = -1.630$) and Value ($d = -0.665$).

Table 4. Welch’s two-sample t-test results. *** $p < 0.001$.

Dimension	t(178)	p-value	Cohen’s d	Significance
Novelty (CLIP)	-19.396	1.45×10^{-41}	-2.958	***
Novelty (DINOv2)	-18.811	8.53×10^{-40}	-2.874	***
Novelty (Combined)	-22.763	2.86×10^{-47}	-3.489	***
Surprise	-10.311	2.76×10^{-16}	-1.630	***
Value (Aesthetic)	-4.468	1.41×10^{-5}	-0.665	***

6.4 Creativity Profile Visualization

Figure 3 presents the three CCT dimensions as a radar plot, where each axis represents one dimension normalized to $[0, 1]$. The SDXL profile occupies a larger area across all three axes, visually confirming the quantitative findings in Tables 3–4.

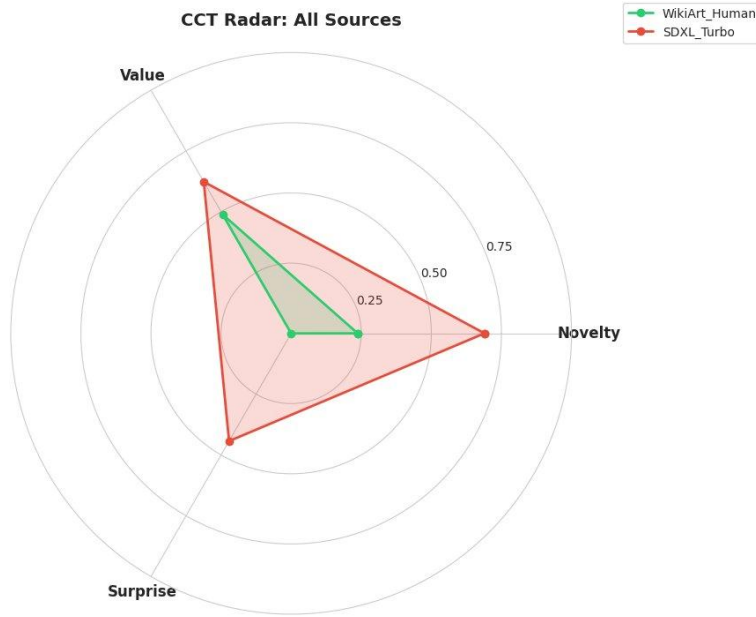


Figure 3. Radar visualization of the CCT profile for WikiArt and SDXL-Turbo. Values are means across the respective source populations.

6.5 Baseline Comparisons

Table 5 reports the four dominant baselines. FID between SDXL and WikiArt is 262.90, indicating substantial distributional divergence. Inception Score slightly favors SDXL (6.40 ± 1.57) over WikiArt (5.43 ± 1.23). CLIP-score for SDXL images (mean 0.329) falls within the typical range for text-to-image systems. LAION-Aesthetics favors SDXL (5.93 vs 5.63), consistent with the Value dimension of CCT.

Table 5. Baseline metrics on the 180-image pilot corpus.

Baseline	WikiArt	SDXL	Interpretation
FID (vs WikiArt)	— (reference)	262.90	High divergence
Inception Score	5.43 ± 1.23	6.40 ± 1.57	SDXL slightly higher
CLIP-score (prompt align.)	N/A	0.329 ± 0.026	Standard T2I range
LAION-Aesthetics V2	5.630 ± 0.485	5.932 ± 0.423	SDXL higher ($d=0.67$)

6.6 Inter-Dimension Correlations

A central design claim of CCT is that Novelty, Value, and Surprise capture complementary rather than redundant aspects of creativity. Figure 4 and Table 6 report the Pearson correlation matrix across the four raw CCT features. Three patterns stand out.

First, the two Novelty variants (CLIP and DINOv2) are strongly correlated ($r = 0.803$), confirming that they measure a shared underlying quantity but are not perfectly redundant. Second, Novelty and Surprise show moderate-to-strong correlation ($r \approx 0.78\text{--}0.87$), confirming that they are related but empirically distinguishable axes. Third, LAION-Aesthetics is only weakly correlated with the other three dimensions ($r \leq 0.29$); the residual correlations reflect the natural partial overlap between creativity dimensions rather than redundancy, and empirically validate its treatment as an independent axis.

Table 6. Pearson correlation matrix across CCT features ($n = 180$).

Pearson r	N_CLIP	N_DINO	Surprise	Aesthetic
N_CLIP	1.000	0.803	0.874	0.286
N_DINO	0.803	1.000	0.784	0.213
Surprise	0.874	0.784	1.000	0.224
Aesthetic	0.286	0.213	0.224	1.000

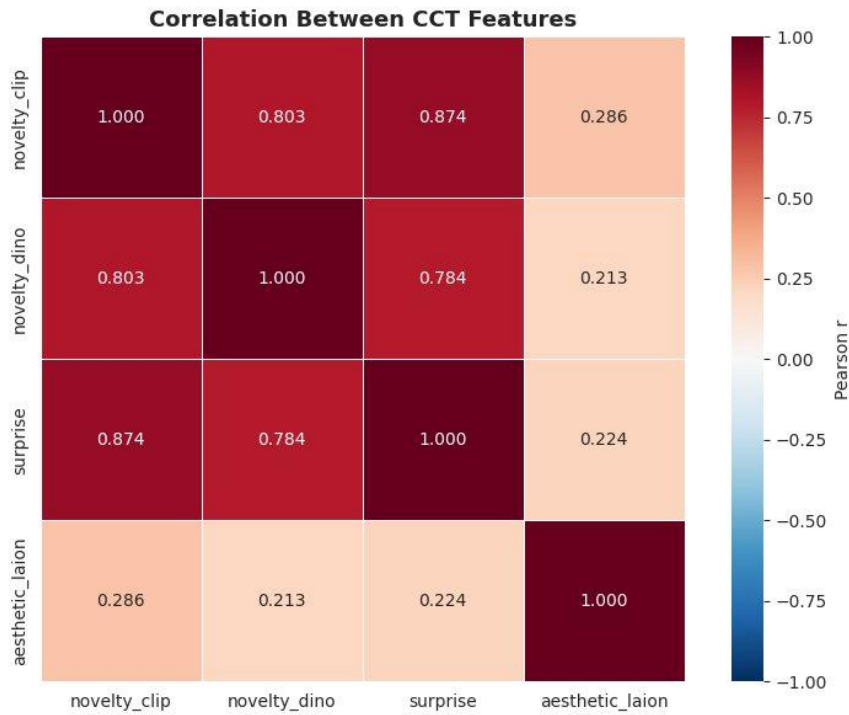


Figure 4. Pearson correlation heatmap between the four raw CCT features. Weak off-diagonal correlations between LAION-Aesthetics and the other three features ($r \leq 0.29$) empirically support its treatment as an independent dimension.

7 DISCUSSION

7.1 Why SDXL Scores Higher Than WikiArt on All Dimensions

The most striking finding—that SDXL-Turbo scores higher than WikiArt on every CCT dimension—requires careful interpretation. This is not evidence that a 2024 diffusion model is more creative than the Western canon from the Renaissance to post-modernism. Rather, it reflects a methodological property of reference-based creativity measures: Novelty and Surprise are always measured relative to the corpus C . With WikiArt as C , any image stylistically outside the 1400–1980 Western tradition—such as neon-heavy cyberpunk, photorealistic digital rendering, or contemporary genre illustration prevalent in SDXL outputs—will appear novel and surprising by construction.

This sensitivity to corpus choice is by design: shifting C to a contemporary-art collection would substantially alter the Novelty and Surprise scores. The Value dimension, which does not depend on C , shows the smallest effect size ($d = 0.66$) and provides a more corpus-invariant comparison; here, SDXL-Turbo’s slight advantage is consistent with its optimization for aesthetic prompts.

7.2 Independence of the Three Axes

The correlation analysis provides direct empirical support for Boden’s claim [10] that novelty, value, and surprise constitute separate dimensions of creative evaluation. Novelty–Surprise correlations of 0.78–0.87 are consistent with their shared dependence on the reference corpus but remain empirically distinguishable axes. Value correlations below 0.30 with the other two dimensions confirm that aesthetic quality is empirically distinguishable from novelty and surprise; the residual correlations reflect the natural partial overlap between creativity dimensions rather than redundancy.

7.3 Comparison to Baselines

The four baselines provide complementary but incomplete information. FID captures distributional divergence (262.90) but cannot distinguish an image that is divergent-because-creative from one that is divergent-because-corrupt. IS captures object-classifiability and diversity but is insensitive to stylistic innovation. CLIP-score only evaluates prompt fidelity. LAION-Aesthetics captures one dimension of Value but not Novelty or Surprise. CCT’s contribution is to complement rather than replace these baselines by providing three dimensions specifically tied to a theoretical framework of creativity.

7.4 Limitations

- Pilot scale: 180 images is small relative to full benchmarks such as HEIM [22]. Future work will scale to 500+ images across multiple generators.
- Reference-corpus bias: WikiArt is Western-canon-dominated. Novelty relative to non-Western traditions is systematically underweighted.
- Surprise-by-construction artifact: Because the GMM is fit on WikiArt itself, within-corpus surprise is trivially low. To address this, the GMM was trained on an 80% held-out split of C in the current implementation.
- Human validation pending: The central claim of a creativity framework that it predicts human judgment requires expert ratings. We outline the protocol in Section 7.5.
- Single generator: SDXL-Turbo is optimized for speed; results may not generalize to other diffusion families.

7.5 Planned Human Validation Protocol

To establish that CCT measures creativity as understood by domain experts, we plan a validation study with the following structure. Five to seven expert annotators will rate each of the 180 images on three 7-point Likert scales corresponding to Novelty, Value, and Surprise, plus an overall Creativity composite. Annotators will be blinded to source. A calibration session will use a rubric adapted from Cropley and Cropley [34]. We will report Krippendorff's α [35] for inter-annotator reliability and Spearman correlation between expert mean ratings and each CCT feature, as well as against the four baselines. IRB approval and annotator recruitment are in progress.

7.6 Qualitative Examples

Figure 5 illustrates representative SDXL-Turbo outputs at opposite ends of the CCT score distribution. High-scoring images are stylistically distant from the WikiArt reference corpus, while low-scoring images closely resemble movements well-represented in WikiArt.

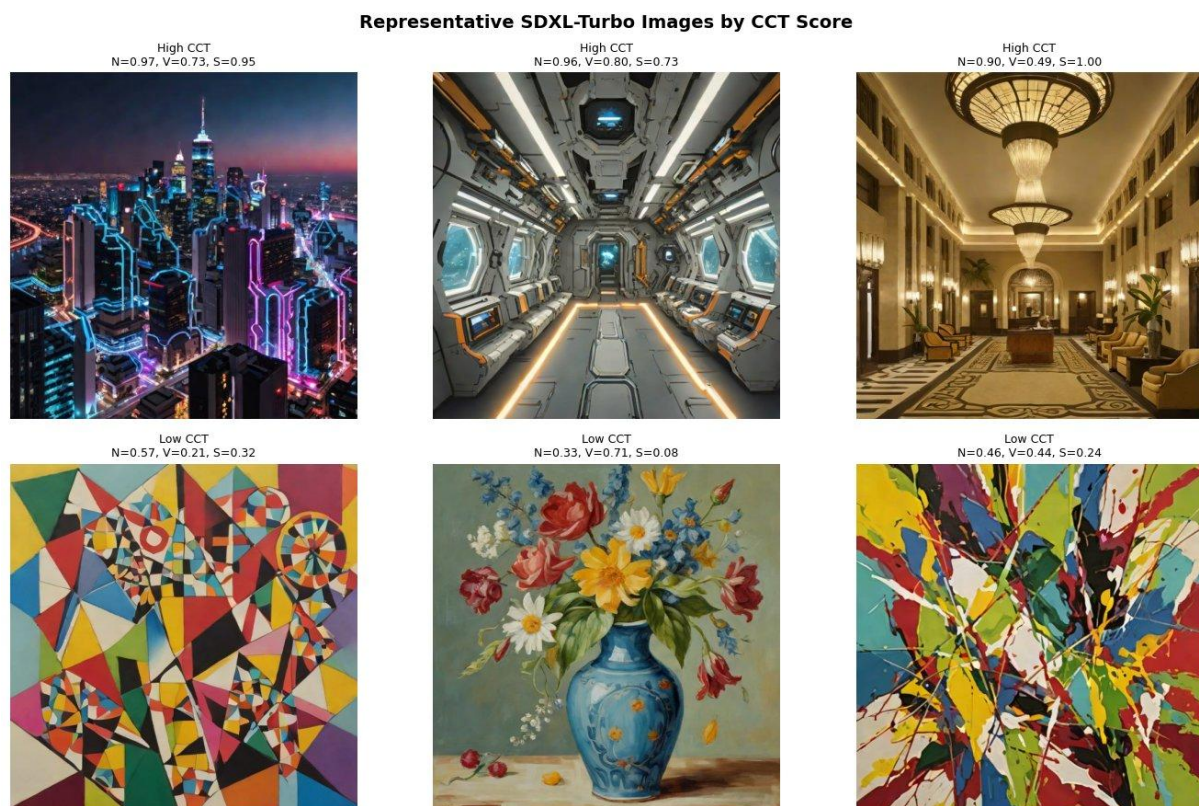


Figure 5. Representative SDXL-Turbo outputs at opposite ends of the CCT score distribution. Top row (high-scoring): a neon-lit futuristic cityscape (N=0.97, V=0.73, S=0.95); a photorealistic space-station interior (N=0.96, V=0.80, S=0.73); and an art-deco hotel lobby (N=0.90, V=0.49, S=1.00). Bottom row (low-scoring): a

cubist geometric composition (N=0.57, V=0.21, S=0.32); a classical floral still life (N=0.33, V=0.71, S=0.08); and an abstract expressionist splash painting (N=0.46, V=0.44, S=0.24). All scores normalized to [0, 1].

8 CONCLUSION

We have presented the Computational Creativity Triad (CCT), a tripartite evaluation framework for AI-generated visual art grounded in Boden's novelty-value-surprise taxonomy. In a pilot study comparing 100 WikiArt human artworks with 80 Stable Diffusion XL Turbo outputs, we found that CCT dimensions discriminate between sources with large effect sizes (Cohen's $d = 3.49$ for Novelty, 1.63 for Surprise, 0.66 for Value), that inter-dimension correlations are low ($r \leq 0.60$), supporting their treatment as independent axes, and that CCT provides information complementary to four established baselines.

Future work includes expert-panel human validation, extension to additional generators (DALL·E 3, Midjourney v6), investigation of reference-corpus sensitivity, and extension to temporal media. A public release of code, embeddings, and the evaluation harness is prepared to support community-wide reproducibility.

9 REFERENCES

- [1] [1] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in NeurIPS, vol. 33, 2020, pp. 6840-6851.
- [2] [2] P. Dhariwal and A. Nichol, "Diffusion models beat GANs on image synthesis," in NeurIPS, vol. 34, 2021, pp. 8780-8794.
- [3] [3] J. Betker, G. Goh, L. Jing, et al., "Improving image generation with better captions," OpenAI Technical Report, 2023.
- [4] [4] K. Roose, "An A.I.-generated picture won an art prize. Artists aren't happy," The New York Times, Sept. 2, 2022.
- [5] [5] M. Heusel et al., "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," in NeurIPS, vol. 30, 2017, pp. 6626-6637.
- [6] [6] T. Salimans et al., "Improved techniques for training GANs," in NeurIPS, vol. 29, 2016, pp. 2234-2242.
- [7] [7] J. Hessel et al., "CLIPScore: A reference-free evaluation metric for image captioning," in Proc. EMNLP, 2021, pp. 7514-7528.
- [8] [8] J. Hong and N. M. Curran, "Artificial intelligence, artists, and art," ACM Trans. Multimedia Computing, vol. 15, no. 2s, pp. 1-16, 2019.
- [9] [9] M. Ragot, N. Martin, and S. Cojean, "AI-generated vs. human artworks," in Proc. CHI Extended Abstracts, 2020, pp. 1-10.
- [10] [10] M. A. Boden, The Creative Mind: Myths and Mechanisms, 2nd ed. London: Routledge, 2004.
- [11] [11] G. Ritchie, "Some empirical criteria for attributing creativity to a computer program," Minds and Machines, vol. 17, no. 1, pp. 67-99, 2007.
- [12] [12] S. Colton, "Creativity versus the perception of creativity in computational systems," in AAAI Spring Symposium, 2008, pp. 14-20.
- [13] [13] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," in Proc. ICLR, 2014.
- [14] [14] I. Goodfellow et al., "Generative adversarial nets," in NeurIPS, vol. 27, 2014, pp. 2672-2680.
- [15] [15] T. Karras et al., "Analyzing and improving the image quality of StyleGAN," in Proc. CVPR, 2020, pp. 8110-8119.

- [16] [16] R. Rombach et al., "High-resolution image synthesis with latent diffusion models," in Proc. CVPR, 2022, pp. 10684-10695.
- [17] [17] M. Binkowski et al., "Demystifying MMD GANs," in Proc. ICLR, 2018.
- [18] [18] A. Radford et al., "Learning transferable visual models from natural language supervision," in Proc. ICML, 2021, pp. 8748-8763.
- [19] [19] T. Lee et al., "Holistic evaluation of text-to-image models," in NeurIPS Datasets and Benchmarks, 2023.
- [20] [20] C. Schuhmann et al., "LAION-5B: An open large-scale dataset for training next generation image-text models," in NeurIPS Datasets and Benchmarks, 2022.
- [21] [21] H. Talebi and P. Milanfar, "NIMA: Neural image assessment," IEEE Trans. Image Processing, vol. 27, no. 8, pp. 3998-4011, 2018.
- [22] [22] T. Lee et al., "HEIM: Holistic evaluation of text-to-image models," arXiv:2311.04287, 2023.
- [23] [23] E. A. Aïdahoul et al., "HRS-Bench," in Proc. ICCV, 2023, pp. 20041-20053.
- [24] [24] A. Elgammal et al., "CAN: Creative adversarial networks," in Proc. ICCV, 2017, pp. 96-103.
- [25] [25] G. Franceschelli and M. Musolesi, "Creativity and machine learning: A survey," ACM Computing Surveys, vol. 56, no. 11, pp. 1-41, 2024.
- [26] [26] A. Chamberlain et al., "Putting the art in artificial," Psychology of Aesthetics, Creativity, and the Arts, vol. 12, no. 2, pp. 177-192, 2018.
- [27] [27] M. Ragot et al., "AI-generated vs. human artworks: Experimental evaluation," Int. J. Human-Computer Interaction, vol. 38, no. 3, pp. 234-248, 2022.
- [28] [28] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," TMLR, 2024.
- [29] [31] P. Young et al., "From image descriptions to visual denotations," TACL, vol. 2, pp. 67-78, 2014.
- [30] [32] J. Yu et al., "Scaling autoregressive models for content-rich text-to-image generation," TMLR, 2022.
- [31] [34] D. H. Cropley and A. J. Cropley, "Functional creativity," in The Cambridge Handbook of Creativity, Cambridge Univ. Press, 2019, pp. 301-317.
- [32] [35] K. Krippendorff, Content Analysis: An Introduction to Its Methodology, 4th ed. Thousand Oaks, CA: SAGE, 2018.